



Maritime &
Coastguard
Agency

Guidance

MGN 301 Amendment 1: Appendix 2

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Manoeuvring Booklet

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Appendix 2 - Recommended Information to be Included in the Manoeuvring Booklet

1. General Description

1.1 Ship's particulars

1.1.1 General

- Ship's name, distinctive number or letters, year of build

1.1.2 Gross tonnage and other information.

- Gross tonnage, dead-weight and displacement (at summer draught)

1.1.3 Principal dimensions and coefficients.

- Length overall, length between perpendiculars, breadth (moulded), depth (moulded), summer draught, normal ballast draught, hull coefficients at summer load and normal ballast condition
- Extreme height of the ship's structure above the keel

1.1.4 Main engine.

- Type, number of units and power output

1.1.5 Propeller.

- Type, number of units, diameter, pitch, direction of rotation, propeller immersion

1.1.6 Rudder.

- Type, number of units, total rudder area, rudder area ratio (full load and normal ballast)

1.1.7 Bow and stern thrusters.

- Type, number of units, capacities and location

1.1.8 Bow and stern profiles.

1.1.9 Forward and after blind zones with dimensions specified (full load and normal ballast).

1.1.10 Other hull particulars.

- Projected areas of longitudinal and lateral above-water profiles (full load and normal ballast)
- Length of parallel middle body for berthing (full load and normal ballast)

1.2 Characteristics of main engine

1.2.1 Manoeuvring speed tables.

- trial or estimated, at the full load and ballast conditions
- Engine revolutions, ship speed and thrust (at ahead) corresponding to engine orders

1.2.2 Critical revolutions.

1.2.3 Time for effecting changes in engine telegraph settings as in 3.1.2 for both routine and emergency conditions.

1.2.4 Tune limit astern.

1.2.5 Minimum operating revolutions (for diesel engines) and corresponding ship speed.

1.2.6 Maximum number of consecutive starts (for diesel engines).

2. Manoeuvring Characteristics in Deep Water

2.1 Course change performance

2.1.1 Initial turning test results (trial or estimated at the full load and ballast conditions), test conditions, diagrams of heading angle versus time and ship's track.

2.1.2 Course change test results (trial or estimated, at full load and ballast conditions) Curves of course change distance and point of initiation of counter rudder for the necessary course change angle (for both full load and ballast conditions).

2.2 Turning circles in deep water (trial or estimated, at the full load and ballast conditions)

2.2.1 Turning circle test results

- Test conditions, test results (advance and transfer) and turning track at full sea speed ahead

2.2.1.1 Turning circles in both full load and ballast conditions (stern track should be shown).

2.2.1.2 The data presented should refer to the case of starboard turn only (unless there is significant difference for port turn).

2.2.1.3 The initial speed of the ship should be full sea speed ahead.

2.2.1.4 Times and speeds at 90°, 180°, 270° and 360° turning should be specifically shown together with an outline of the ship.

2.2.1.5 The rudder angle used in the test should be the maximum rudder angle.

2.3 Accelerating turn (trial or estimated)

2.3.1 Data are to be presented for both full load and ballast conditions in the same manner as 2.2 for turning circles. The ship accelerates from rest with the engine full manoeuvring speed ahead and the maximum rudder angle.

2.4 Yaw checking tests (trial or estimated)

2.4.1 Results of the zigzag and pull-out manoeuvre tests at the full load or ballast condition shown as diagrams of the heading changes and rudder angle.

2.5 Man-overboard and parallel course manoeuvres

2.5.1 Man-overboard manoeuvre (trial).

- Diagrams for cases of both starboard and port turns should be shown for both full load and ballast conditions

2.5.2 Parallel course manoeuvre (estimated).

- Diagrams showing lateral shift to a parallel course using maximum rudder angle

2.6 Lateral thruster capabilities (trial or estimated)

2.6.1 Diagrams of turning performance at zero forward speed in the full load or ballast condition should be shown, for bow and stern thrusters acting separately and in combination.

2.6.2 Diagrams showing the effect of forward speed on turning performance should be included.

2.6.3 Information on the effect of wind on turning performance should be given.

3. Stopping and Speed Control Characteristics in Deep Water

3.1 Stopping ability

3.1.1 Stopping test results (trial).

- Test conditions, ship's tracks, rpm, speed, track reach, head reach and side reach
- Two or more tests should be carried out including a test of full astern from full sea speed ahead and a test of full astern from full ahead speed.

3.1.2 Stopping ability (estimated)

Information and diagrams should be given of the track reach, head reach, side reach, time required and track reach deceleration factor (distance/one knot reduction) of a ship in both full load and ballast conditions covering the following modes of stopping manoeuvres:

- full astern from full sea speed ahead
- full astern from full ahead speed
- full astern from half ahead speed
- full astern from slow ahead speed
- stop engine from full sea speed ahead
- stop engine from full ahead speed
- stop engine from half ahead
- speed stop engine from slow ahead speed

3.2 Deceleration performance (estimated)

3.2.1 Deceleration ability (estimated).

Information and diagrams should be given concerning the track reach, time required and deceleration factor of the ship in both full load and ballast conditions for the following engine orders:

- full sea speed to “stand by engines”
- full ahead to half ahead
- half ahead to slow ahead
- slow ahead to dead slow ahead

3.3 Acceleration performance (estimated)

3.3.1 Information and diagrams should be given for track reach and time for the ship to achieve full sea speed ahead, from zero speed.

4. Manoeuvring Characteristics in Shallow Water

4.1 Turning circle in shallow water (estimated)

4.1.1 Turning circle in the full load condition (stern track to be shown).

4.1.2 The initial speed of the ship should be half ahead.

4.1.3 Times and speeds at 90°, 180°, 270° and 360° turning should be specifically shown, together with an outline of the ship.

4.1.4 The rudder angle should be the maximum and the water depth to draught ratio should be 1.2.

4.2 Squat (estimated)

4.2.1 Curves should be drawn for shallow water and infinite width of channel, indicating the maximum squat versus ship speed for various water depth/draught ratios.

4.2.2 Curves should be drawn for shallow and confined water, indicating the maximum squat versus speed for different blockage factors.

5. Manoeuvring Characteristics in Wind

5.1 Wind forces and moments (estimated)

5.1.1 Information should be given on the wind forces and moments acting on the ship for different relative wind speeds and directions in both full load and ballast conditions, to assist in berthing.

5.2 Course-keeping-limitation (estimated)

5.2.1 Information should be given for both full load and ballast conditions, showing the effect of wind on the ability of the ship to maintain its course.

5.3 Drifting under wind influence (estimated)

5.3.1 Information should be given on the drifting behaviour under wind influence with no engine power available.

6. Manoeuvring Characteristics at Low Speed (Trial or Estimated)

6.1 Information on the minimum operating revolutions of the main engine and corresponding ship's speed should be given.

6.2 Information on the minimum speed at which the ship can maintain course while still making headway after stopping engines.

7. Additional Information

7.1 Any other relevant additional information should be added to the contents of the booklet, particularly information concerned with the operation of the bridge manoeuvring controls.



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